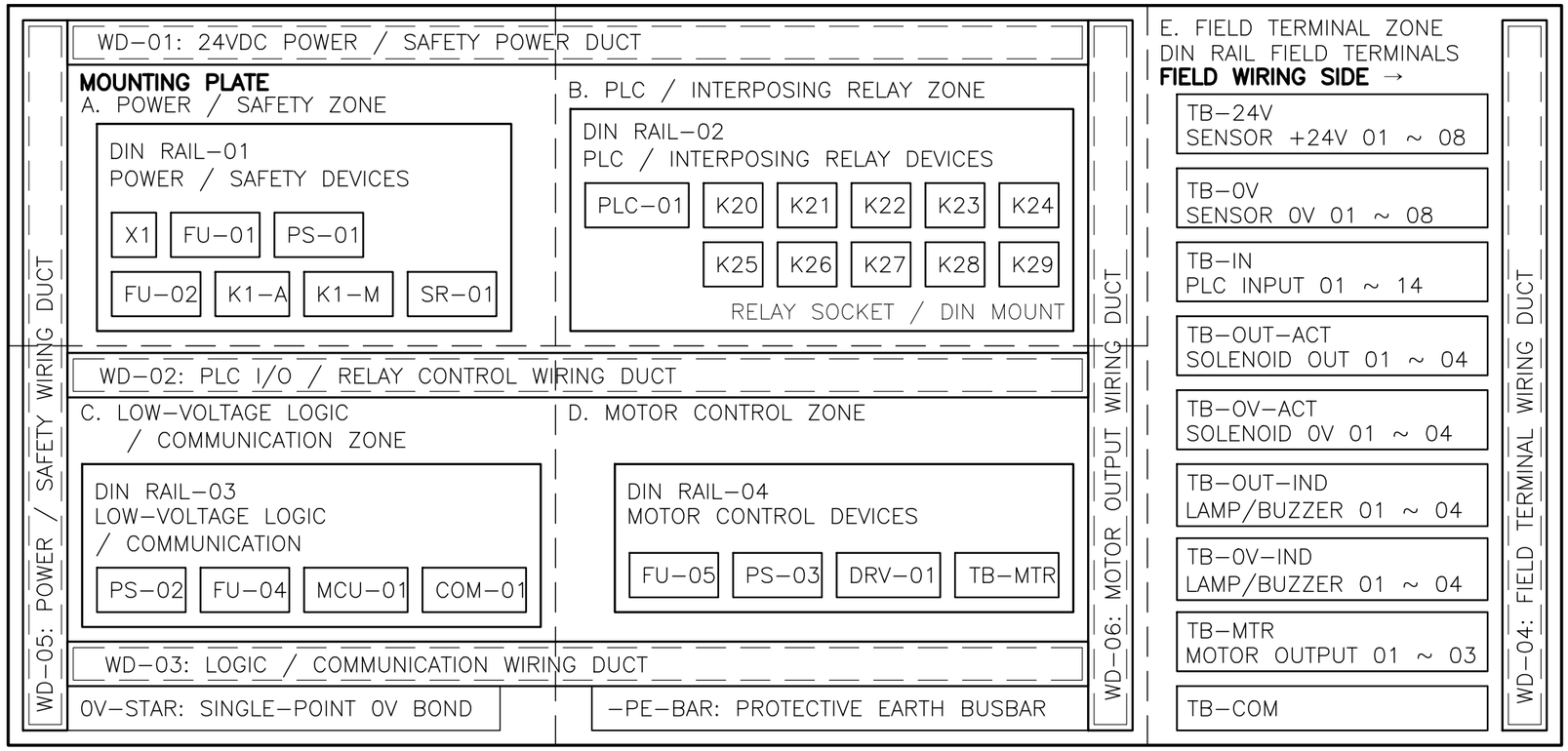
CONTROL PANEL INTERNAL LAYOUT CONCEPTUAL ARRANGEMENT – NOT TO SCALE



[RELAY]

K20
MTR RUN CMD

K21
MTR DIR CMD

K22
SOL–01 CLAMP

K23
SOL–02 PASS

K24
SOL–03 FAIL

K25
SOL–04 RETEST

K26
LT–01 GREEN

K27
LT–01 YELLOW

K28
LT–01 RED

K29
BZ–01 BUZZER

PANEL COMPONENT LEGEND		
Symbol / Tag	Description	Reference
PS-01	24VDC Control Power Supply	DWG 005
PS-02	5VDC Logic Power Supply	DWG 005
PS-03	Motor Power Supply	DWG 005
SR-01	Safety Relay	DWG 006
PLC-01	LS XBC-DR32H Controller	DWG 004 / 007 / 008
K1-A	Safety Actuator Power Relay	DWG 005 / 006
K1-M	Motor Safety Power Relay	DWG 005 / 006
K20~K29	PLC Output Interposing Relays	DWG 008
MCU-01	ESP32-S3 Controller	DWG 003 / 008
COM-01	TTL to RS-485 Interface	DWG 003
DRV-01	DC Motor Driver	DWG 008
TB-*	Field Terminal Block Group	DWG 007 / 008
X1	Main Incoming Power / PE Terminal	DWG 005
FU-01	Control Power Fuse	DWG 005
FU-02	Safety Actuator Power Fuse	DWG 005
FU-04	Logic Power Fuse	DWG 005
FU-05	Motor Power Fuse	DWG 005

0V–STAR:
0V_CTRL / 0V_LOGIC / 0V_MOTOR
BONDED AT SINGLE–POINT STAR GROUND.

–PE–BAR:
PANEL ENCLOSURE
MTR–01 FRAME
CABLE SHIELD
X1–PE

TB–COM:
RS–485 A / B / GND
01 ~ 03

[GENERAL NOTES]

[1] THIS DRAWING SHOWS CONCEPTUAL CONTROL PANEL INTERNAL COMPONENT ARRANGEMENT. FINAL PANEL DIMENSIONS SHALL BE VERIFIED FROM ACTUAL DEVICE DATA SHEETS.

[2] AC / 24VDC POWER, SAFETY, PLC I/O, LOGIC COMMUNICATION, AND MOTOR POWER WIRING SHALL BE ROUTED IN SEPARATE DUCTS.

[3] MCU–01, COM–01, AND RS–485 WIRING SHALL BE LOCATED AWAY FROM DRV–01, TB–MTR, AND MOTOR OUTPUT WIRING.

[4] FIELD TERMINAL BLOCKS SHALL BE LOCATED ON THE FIELD WIRING SIDE OF THE CONTROL PANEL.

[5] +24V_SAFE_ACT SHALL BE CONTROLLED THROUGH K1–A. +V_MOTOR_SAFE SHALL BE CONTROLLED THROUGH K1–M.

[6] PE–BAR SHALL BE CONNECTED TO PANEL ENCLOSURE, X1–PE, MTR–01 FRAME, AND APPLICABLE CABLE SHIELDS.

[7] 0V_CTRL, 0V_LOGIC, AND 0V_MOTOR SHALL BE BONDED ONLY AT THE DESIGNATED SINGLE–POINT 0V STAR GROUND.

[8] FINAL PANEL FABRICATION SHALL PROVIDE A MINIMUM 20 PERCENT SPARE DIN RAIL AND WIRE DUCT CAPACITY FOR FUTURE EXPANSION.

[9] VERIFY FINAL DEVICE DIMENSIONS, HEAT DISSIPATION, CLEARANCE, TERMINAL RATING, AND WIRE BENDING RADIUS BEFORE PANEL FABRICATION.

NO.	DATE	REVISION
A	2026–09–02	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi–Bin Sorting Cell

PANEL MOUNTING PLATE / DIN RAIL / WIRE DUCT ARRANGEMENT

DRAWN BY	DATE
J. H. KO	2026–09–02

DWG NO
009

SHEET NO
1 OF 2